Helena Crosby

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Education

University of Central Florida — Orlando, FL
Bachelor of Science in Computer Science — *Aug. 2026 - Jul. 2029*

Experience

Software Engineer Intern — Feb. 2026 - Aug. 2026
MascotGO — *Remote*

- Wrote parameterized SQL joins to rank related schools by relationship strength, returning results through a recommendations endpoint.
- Designed PostgreSQL schema for tracking peer, competitor, and aspirational school relationships, with source attribution, foreign keys, and CHECK constraints for data integrity

Mentor — Feb. 2026 – Mar. 2026
ReDesignHer — *Remote*

- Mentored students in a 4-week virtual program focused on design thinking, creative problem-solving, and innovation.
- Provided weekly guidance and feedback as students developed solutions to real-world challenges.

Projects

Omnibelt | *Swift, ARKit, Computer Vision, NVIDIA Jetson Orin Nano*

- Developed an iOS application using Swift and Xcode for an AI-powered wearable designed to assist blind and low-vision individuals with navigation.
- Integrated Apple's ARKit and computer vision capabilities to detect and store information about surrounding objects.
- Implemented Wi-Fi communication between the iOS application and an NVIDIA Jetson Orin Nano, connecting mobile software with embedded hardware.
- Collaborated with a team to develop an award-winning assistive technology prototype during Knight Hacks IX.

Setlistics | *Next.js, React, Rest API, Vercel*

- Built and deployed a full-stack Next.js app that pulls live data from the setlist.fm and iTunes Search APIs to rank an artist's most-played songs and cities.
- Wrote server-side API routes to keep credentials private, and added caching, retries, and request pacing to work within API rate limits.
- Implemented a tour filter, album art, and 30-second song previews, and built a responsive UI from my own Figma prototype.

Debate Arena | *Python, Flask, Backboard AI, HTML/CSS*

- Architected a multi-agent AI web app where three autonomous agents (PRO, CON, JUDGE) debate and evaluate topics in real time using the Backboard AI API.
- Built a responsive single-page interface with Flask backend integration and light/dark mode theming.

Certifications

AI Fundamentals: Foundations for Understanding AI from IBM

Technical Skills

Languages: Python, Java, SQL, HTML, CSS, C
Frameworks + Libraries: FastAPI, Flask, Next.js, React, Tkinter
Databases: SQLite, PostgreSQL
APIs and AI: Git, GitHub, Docker, Linux, VS Code, Figma
Tools: Git, GitHub, Docker, Linux, VS Code, Figma